

AI Agent Challenge

Housing Loan Arrears Rescheduling

Sheikh Zayed Housing Programme | UAE Ministry of Energy and Infrastructure

1. Brief Introduction about Sheikh Zayed Housing Programme

Sheikh Zayed Housing Programme is a federal housing programme in the United Arab Emirates that supports UAE citizens in achieving housing stability.

The Programme provides housing assistance to eligible citizens through approved policies and procedures. One of the key forms of support is the housing loan, where the beneficiary receives housing financial support and repays it through monthly installments over an approved repayment period.

***Note:** This challenge focuses only on beneficiaries who received housing loans, because they have monthly repayment obligations.*

2. What is Arrears Rescheduling?

Some housing loan beneficiaries may stop paying their monthly installments for a period of time. As a result, unpaid installments accumulate and become **arrears**.

Simple Example

A beneficiary received a housing loan of **AED 800,000**. He was expected to repay the loan through monthly installments. However, he stopped paying for several months, resulting in **AED 40,000 in arrears**.

In this case, the Programme does not only look at the unpaid amount — the beneficiary's overall situation is studied to determine a suitable rescheduling plan.

3. How is the Service Currently Processed?

The beneficiary applies through approved digital channels and mainly submits:

- Salary certificate
- Detailed salary or income statement

- Any supporting documents related to financial or social circumstances

The remaining core data is already available within the Programme:

- Original loan amount
- Remaining loan balance
- Total arrears amount
- Number of unpaid installments
- Remaining repayment period
- Payment history
- Available family or social status data

Key Rule 1	The deduction rate must not exceed 20% of the beneficiary's income.
Key Rule 2	The repayment period must not exceed the original approved loan repayment period .

Currently, this process takes around 5 working days.

4. What is the Current Problem?

The current service depends heavily on manual review, even though much of the required information is structured and available within the Programme.

- Longer service completion time
- Manual workload on employees
- Potential differences in how cases are studied
- The need for more consistent and documented decision-making
- Difficulty in processing clear cases instantly

The objective is to move from a manual process that takes around 5 working days to an instant or near-instant service.

5. What is Expected from the AI Agent?

The challenge is to build an **AI Agent for Housing Loan Arrears Rescheduling**. This should not be a normal chatbot — the Agent should act as a smart assistant that studies each rescheduling request in a structured way, similar to how an employee reviews the case.

The Agent should analyze:

- Beneficiary income
- Arrears amount
- Remaining loan balance
- Remaining repayment period
- Number of unpaid installments
- Payment history
- Marital and family status
- Number of family members
- Average income per family member
- Whether the income is stable or has changed
- Supporting documents submitted by the beneficiary
- The beneficiary's realistic repayment capacity

6. Why Should the Agent Study Each Case Differently?

Not all beneficiaries should receive the same rescheduling plan.

Example 1	Example 2
A married beneficiary with no children, stable income, and a good average income per family member may have a higher repayment capacity .	A married beneficiary with several children, a reduction in income, and an average income per family member below AED 2,500 may require a different, lighter repayment plan .

7. What Should the AI Agent Do?

1	Read the rescheduling request.
2	Verify that the salary certificate, income statement, and supporting documents are available.
3	Retrieve the available Programme data related to the loan, arrears, and payment history.
4	Analyze the beneficiary's income, family status, and social situation.
5	Calculate the average income per family member when needed.
6	Analyze the arrears amount, remaining loan balance, and remaining repayment period.
7	Test whether rescheduling is possible without exceeding 20% of the beneficiary's income.
8	Ensure that the proposed plan does not exceed the original loan repayment period.

9	Recommend a suitable rescheduling plan based on the case.
10	Explain the reasoning behind the recommendation.
11	Refer the case to a human employee if it is complex or requires human judgment.

8. Expected Output from the AI Agent

The AI Agent should provide a clear and structured recommendation including:

Application Status	Complete / Incomplete
Case Summary	Overview of the beneficiary's situation
Income Analysis	Salary, stability, per-member average
Arrears Amount	Total unpaid installments
Remaining Loan Balance	Outstanding principal
Remaining Repayment Period	Months/years left
Proposed Deduction Rate	% of income
Proposed Repayment Plan	Duration and installment amount
20% Rule Compliance	Pass / Fail
Period Rule Compliance	Pass / Fail
Recommendation	Approve / Request documents / Refer to employee
Reasoning	Explanation behind the recommendation

9. Simple Summary

Build an AI Agent that studies housing loan arrears rescheduling requests. The Agent takes the beneficiary's submitted documents and the Programme's available data, then analyzes the income, arrears, remaining loan balance, remaining repayment period, family situation, and repayment capacity.

Based on this analysis, the Agent recommends a fair and suitable rescheduling plan, while ensuring that the deduction does not exceed **20% of income** and the repayment period does not exceed the **original loan period**.

Goal: Reduce service time from 5 working days to instant
while ensuring transparency, governance, consistency, and fairness in decision-making